Implementing Tier 2 Interventions for Struggling Readers

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Abstract

Within a Response to Intervention (RTI) model, students may receive tier 2 interventions if they need additional support beyond quality core instruction. The current study was implemented as part of the Path to Reading Excellence in School Sites (PRESS) and examined the effectiveness of standardized tier 2 interventions targeted to struggling readers' specific reading skill deficits. Participants were second and third grade general education students, including 320 identified as struggling readers receiving targeted tier 2 interventions and 411 proficient peers receiving core instruction only over the course of one school year. These results suggested that the reading growth rate of students who received targeted tier 2 interventions was significantly greater than the reading growth rate of their peers who received only core instruction. Potential implications for closing the reading achievement gap will also be discussed.

Introduction

- Response to Intervention (RtI) is a multi-tiered intervention model that allows for interventions to be delivered at different intensities depending on the student's academic needs.
- Tier 2 interventions within an RtI model provide targeted academic or behavioral supports for students for whom quality core instruction (Tier 1) is not sufficient, and includes frequent progress monitoring and small-group instruction (Burns & Coolong-Chaffin, 2006; O'Connor, Harty, & Fulmer, 2005).
- Providing intensive and systematic reading intervention to small groups of struggling readers was the only core component of an RtI model for which there was a strong research base (Gersten et al., 2009), and previous meta-analytic research found moderate effects (*g* = .52) for group interventions (Piasta & Wagner, 2010).
- Vaughn and colleagues (2010) found small effects (*d* = .16) for standardized comprehensive reading interventions at Tier 2.
- Reading development among struggling learners tends to follow a sequence that aligns with the National Reading Panel areas (NRP, 2000): phonemic awareness, phonics, fluency, and vocabulary/comprehension (Berninger et al., 2006).
- Burns and colleagues (Burns, Riley-Tillman, & VanDerHeyden, 2012) suggested that low-level analysis could be used to identify in which NRP category a student struggles in order to match the reading intervention to student need.
- The following research question guided the study: What effect do targeted reading interventions have on the reading growth of struggling second- and third-grade students as compared to the general population?

Method

Participants. Participants for the study were 720 students in second and third grade attending one of the six urban elementary schools participating in the PRESS project. Of the 720 students, 388 (54%) were in second grade and 332 (46%) in third grade; 309 (43%) received a targeted tier 2 intervention and 411 (57%) did not. The total sample consisted of 51.4% females, 14% white students, and 80% were eligible for the Federal Free or Reduced Price Lunch program. Students who received special education services were not included in the analyses.

Measures.

Benchmarking data. Curriculum-based measurement for reading (CBM-R) was measured three times a year (fall, winter, and spring) using AIMSweb passages. Students who were below any benchmark assessment were eligible to receive tier 2 interventions. Data consisted of words read correctly per minute (WRCM) and were converted to a slope metric using ordinary least squares. Thus, the data for the study consisted of average increase in WRCM per week on the benchmark assessment data.

Procedure. Empirically based interventions were delivered to students performing below benchmark. Students received interventions in small groups of 2-3 students for 20-30 minutes. Interventions occurred 3-5 times per week and lasted throughout the entire school year. Interventionists were graduate students who received extensive training on delivering effective interventions. Before interventions began, a wide variety of skill assessments were administered to determine which tier 2 intervention would be most appropriate. The implemented interventions focused on phonics or reading fluency skills, depending on the student need. Interventions came from the PRESS Intervention Manual (available upon request). Treatment fidelity was assessed once each month for each intervention with an intervention checklist. The total number of observed items was divided by the total number of items on the checklist and resulted in at least 90% correct implementation for each assessment.

Analysis. The research question was examined with a two-way analysis of covariance (ANCOVA) in which the slope of the seasonal CBM-R data served as the dependent variable. The two independent variables were a) grade level of the participants, and b) the intervention group (received an intervention – tier 2 or did not – tier 1). Finally, the student's fall CBM-R score served as the covariate for the analysis.

Results

- A one-way ANCOVA was conducted to determine if the school had an effect on student growth as measured by CBM-R slopes. The effect was significant F (5, 713), $p < .05$, but small with a partial eta squared of .03. Therefore, data were collapsed across schools for subsequent analyses.
- Students who received the intervention made significantly more growth F (1,719) = 5.29, $p < .05$ when controlling for the fall CBM-R score. The size of the effect was moderate (2nd grade d = .46, 3rd grade d = .58). The grade level did not have a significant effect on slope value F (1, 719) = .78, p = .38 after controlling for fall CBM-R score, nor did the interaction of receiving a tier 2 intervention and grade F (1, 719) = 3.21, p = .07.

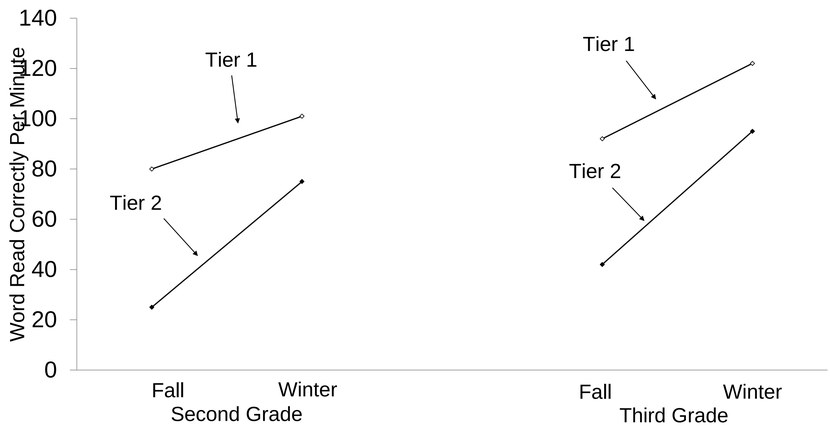


Figure 1. *Seasonal Benchmark Curriculum-Based Measurement Scores for Students Receiving Tier 2 Interventions and Tier 1 Instruction*

Discussion

The current study examined the effects of tier 2 interventions on struggling readers. Results indicated that students who fall below the benchmark, can benefit from small group tier 2 interventions.

Findings

- For those students who are falling below benchmark, and are not demonstrating the appropriate amount of growth in tier 1, tier 2 interventions led to a higher growth rate when compared to students who were not receiving tier 2 interventions. This result is similar to past findings on tier 2 interventions (Burns & Coolong-Chaffin, 2006; O'Connor, Harty, & Fulmer, 2005).
- Although the CBM-R scores for the spring benchmarks were still below the criteria of 90 WRCM for second grade and 109 WRCM for third grade, the gap had narrowed.
- As previous research has found, using data to correctly target the student's need led to interventions that were at the appropriate skill level (Gersten et al., 2008).
- The effects in the current data were moderate (*d* = .46 and .58) and consistent with the Piasta and Wagner (2010) meta-analysis of small-group interventions.

Limitations

- Fidelity checks were only administered once a month, and therefore could have been done more frequently to ensure proper implementation.
- The effects were measured with ORF, which has been shown to link directly to comprehension (Baker et al., 2008), but comprehension was not directly measured.
- The interventions were implemented with trained graduate students.
- The data could have been nested by classroom or school, but they were aggregated across the two conditions. Future research should consider examining the data by appropriate groups and subgroups.

Future Research

- The study focused on students attending inner city schools in a large metropolitan area. Results may differ for schools in rural or suburban areas.
- Future research should consider examining if the same level of growth occurs when the interventions are implemented by school-based personnel.
- Future studies could examine of the effects of length or size of intervention group.
- Finally, future research could directly compare comprehensive reading interventions to those that are directly matched to student deficits.

PRESS
Path to Reading Excellence in School Sites

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